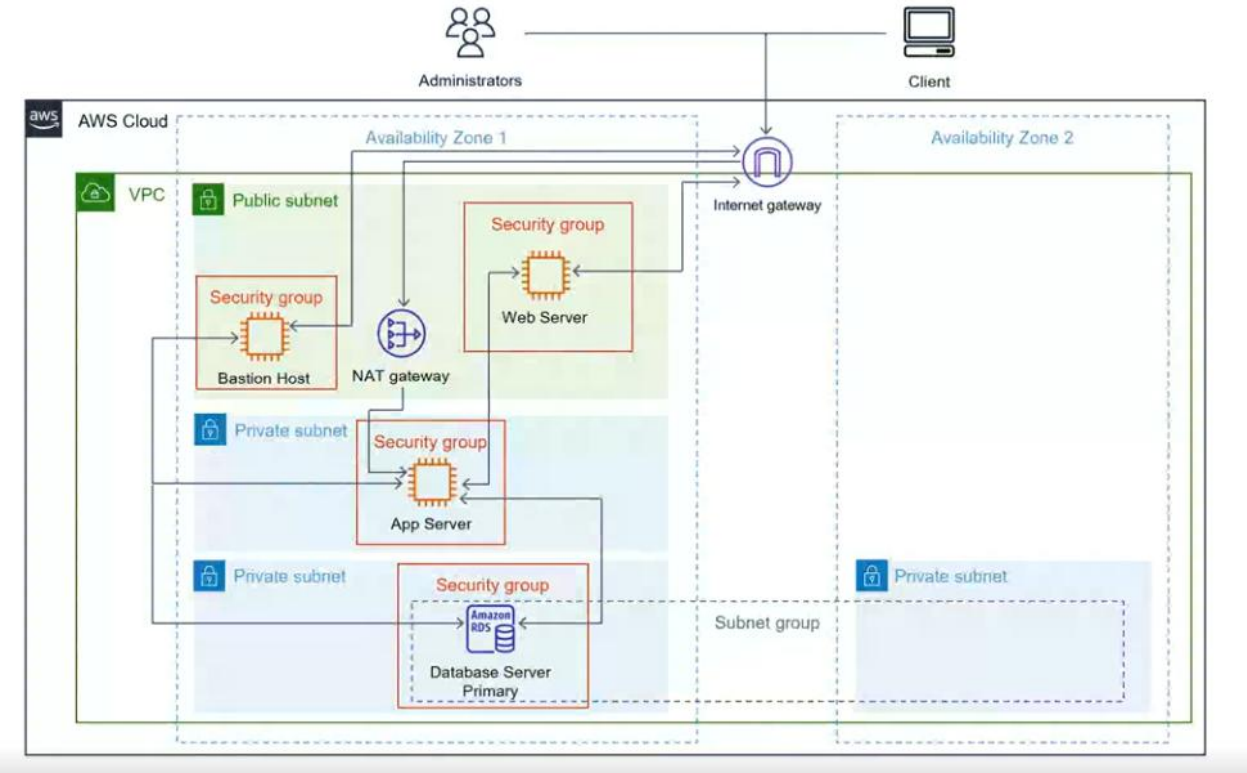


You successfully created vpc-069869e70b5636e3f / vpc-project1

vpc-069869e70b5636e3f / vpc-project1

Actions

Details Info

VPC ID vpc-069869e70b5636e3f	**State** Available	**Block Public Access** Off	**DNS hostnames** Disabled
DNS resolution Enabled	**Tenancy** default	**DHCP option set** dopt-0e1848ac19fc41e7a	**Main route table** rtb-0fae150e2c09e96c8
Main network ACL acl-09bde1fe3572575a4	**Default VPC** No	**IPv4 CIDR** 192.168.0.0/16	**IPv6 pool** –
IPv6 CIDR (Network border group) –	**Network Address Usage metrics** Disabled	**Route 53 Resolver DNS Firewall rule groups** –	**Owner ID** 381491847924
Encryption control ID –	**Encryption control mode** –		

You have successfully created 4 subnets: subnet-09ca36533a97e3599, subnet-00097f08a0eb34079, subnet-0c025139947f24908, subnet-0d0d44c52399e2508

Subnets (8) Info

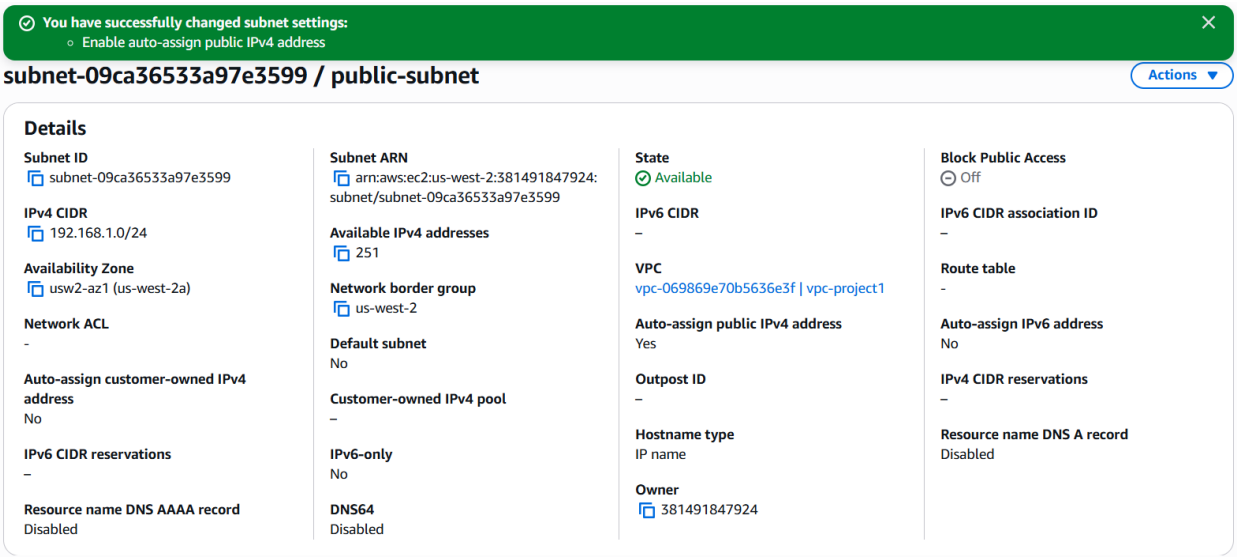
Last updated less than a minute ago | Actions | Create subnet

Find subnets by attribute or tag

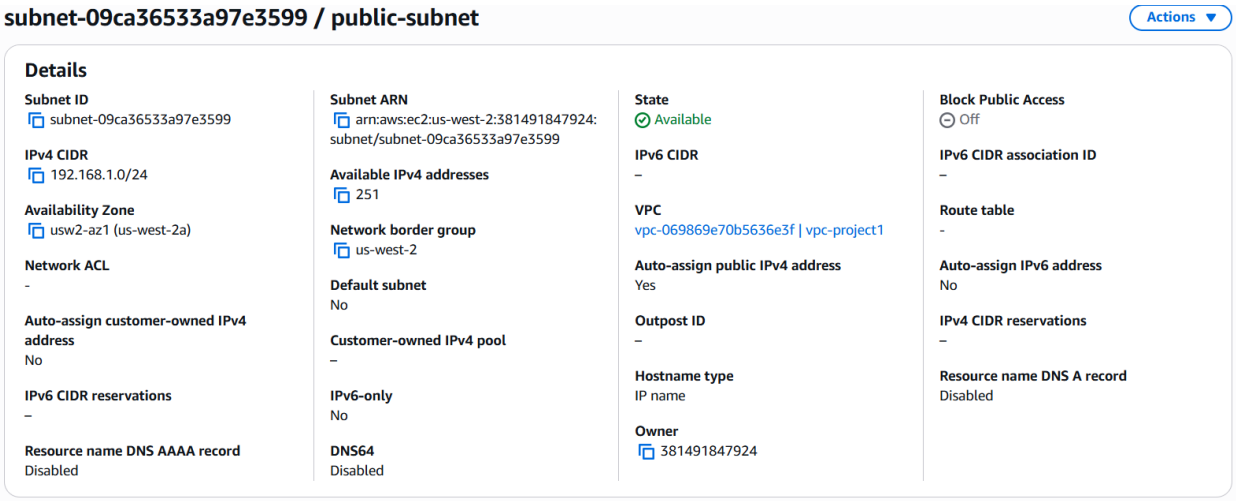
1

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
public-subnet	subnet-09ca36533a97e3599	Available	vpc-069869e70b5636e3f | vpc-project1	Off	192.168.1.0/24
private-subnet3	subnet-0d0d44c52399e2508	Available	vpc-069869e70b5636e3f | vpc-project1	Off	192.168.4.0/24
private-subnet2	subnet-0c025139947f24908	Available	vpc-069869e70b5636e3f | vpc-project1	Off	192.168.3.0/24
private-subnet1	subnet-00097f08a0eb34079	Available	vpc-069869e70b5636e3f | vpc-project1	Off	192.168.2.0/24

Enable in subnet settings public ip addresses



Public subnet, private subnet1 and private subnet2 in same availability zone, private-subnet3 in different availability zone



subnet-00097f08a0eb34079 / private-subnet1

Actions

Details

Subnet ID
subnet-00097f08a0eb34079

IPv4 CIDR
192.168.2.0/24

Availability Zone
usw2-az1 (us-west-2a)

Network ACL
-

Auto-assign customer-owned IPv4 address
No

IPv6 CIDR reservations
–

Resource name DNS AAAA record
Disabled

Subnet ARN
arn:aws:ec2:us-west-2:381491847924:subnet/subnet-00097f08a0eb34079

Available IPv4 addresses
251

Network border group
us-west-2

Default subnet
No

Customer-owned IPv4 pool
–

IPv6-only
No

DNS64
Disabled

State
Available

IPv6 CIDR
–

VPC
vpc-069869e70b5636e3f | vpc-project1

Auto-assign public IPv4 address
No

Outpost ID
–

Hostname type
IP name

Owner
381491847924

Block Public Access
Off

IPv6 CIDR association ID
–

Route table
-

Auto-assign IPv6 address
No

IPv4 CIDR reservations
–

Resource name DNS A record
Disabled

subnet-0c025139947f24908 / private-subnet2

Actions

Details

Subnet ID
subnet-0c025139947f24908

IPv4 CIDR
192.168.3.0/24

Availability Zone
usw2-az1 (us-west-2a)

Network ACL
-

Auto-assign customer-owned IPv4 address
No

IPv6 CIDR reservations
–

Resource name DNS AAAA record
Disabled

Subnet ARN
arn:aws:ec2:us-west-2:381491847924:subnet/subnet-0c025139947f24908

Available IPv4 addresses
251

Network border group
us-west-2

Default subnet
No

Customer-owned IPv4 pool
–

IPv6-only
No

DNS64
Disabled

State
Available

IPv6 CIDR
–

VPC
vpc-069869e70b5636e3f | vpc-project1

Auto-assign public IPv4 address
No

Outpost ID
–

Hostname type
IP name

Owner
381491847924

Block Public Access
Off

IPv6 CIDR association ID
–

Route table
-

Auto-assign IPv6 address
No

IPv4 CIDR reservations
–

Resource name DNS A record
Disabled

subnet-0d0d44c52399e2508 / private-subnet3

Actions

Details

Subnet ID
subnet-0d0d44c52399e2508

IPv4 CIDR
192.168.4.0/24

Availability Zone
usw2-az2 (us-west-2b)

Network ACL
-

Auto-assign customer-owned IPv4 address
No

IPv6 CIDR reservations
–

Resource name DNS AAAA record
Disabled

Subnet ARN
arn:aws:ec2:us-west-2:381491847924:subnet/subnet-0d0d44c52399e2508

Available IPv4 addresses
251

Network border group
us-west-2

Default subnet
No

Customer-owned IPv4 pool
–

IPv6-only
No

DNS64
Disabled

State
Available

IPv6 CIDR
–

VPC
vpc-069869e70b5636e3f | vpc-project1

Auto-assign public IPv4 address
No

Outpost ID
–

Hostname type
IP name

Owner
381491847924

Block Public Access
Off

IPv6 CIDR association ID
–

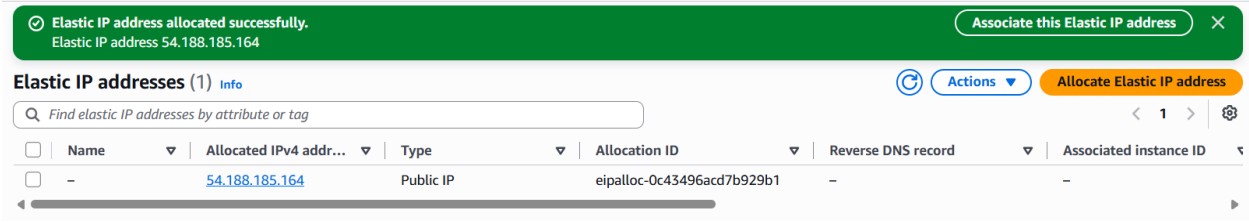
Route table
-

Auto-assign IPv6 address
No

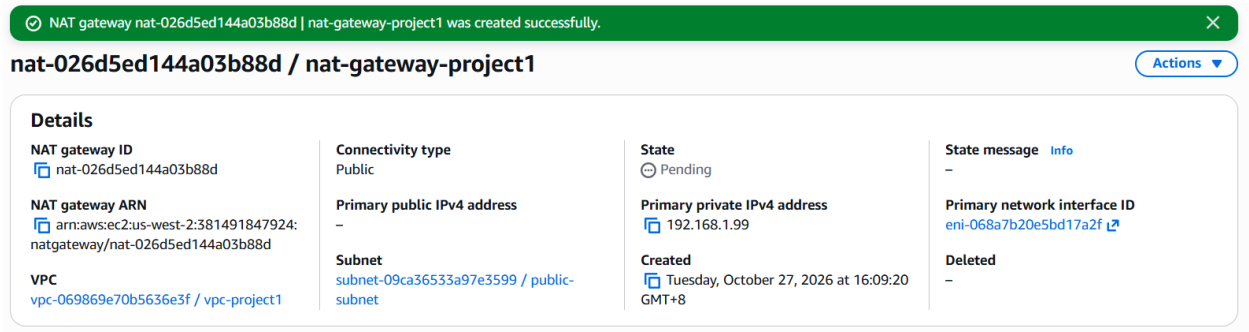
IPv4 CIDR reservations
–

Resource name DNS A record
Disabled

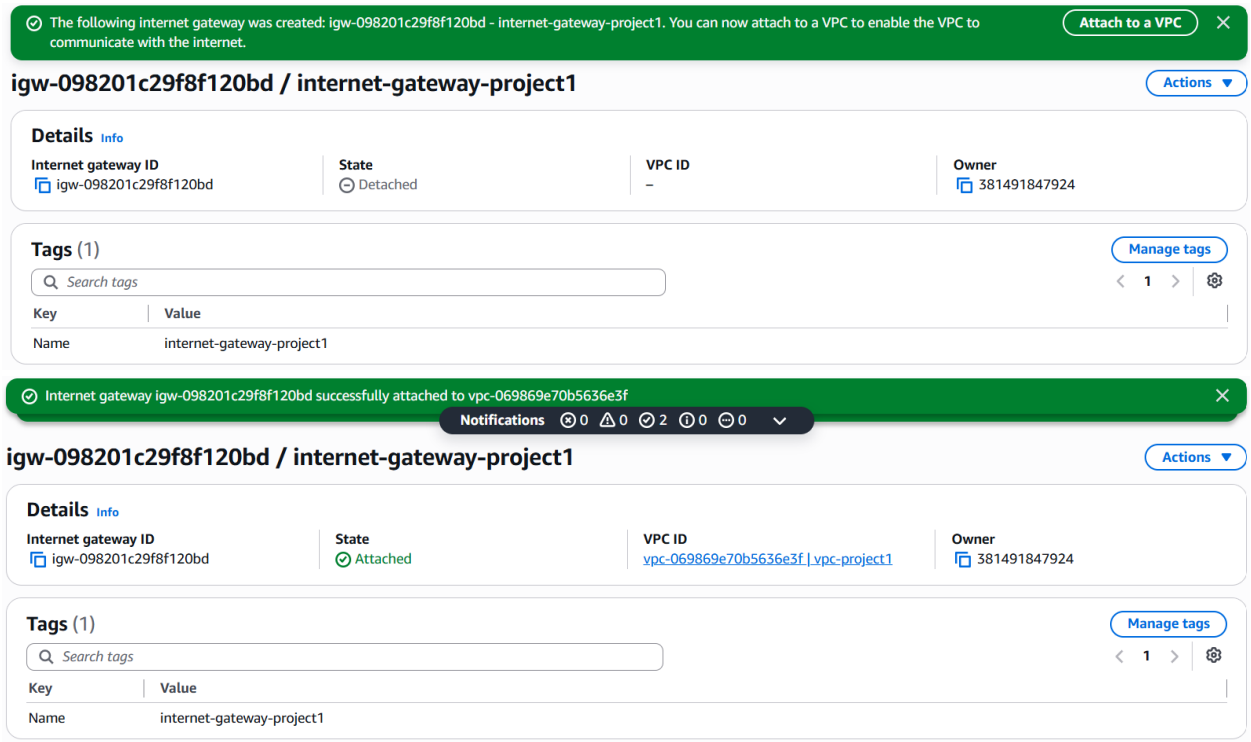
Allocate Elastic IP



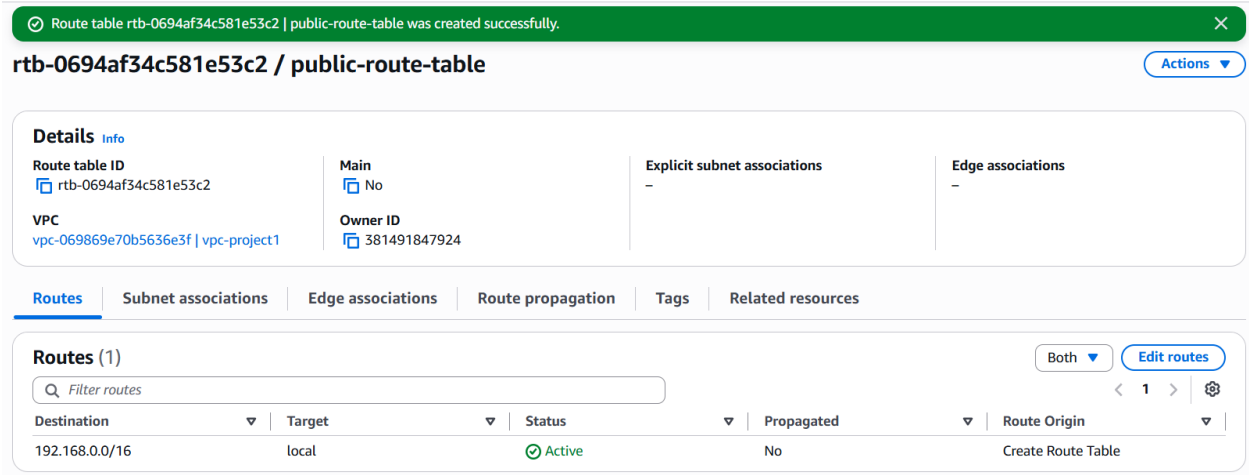
Create NAT gateway



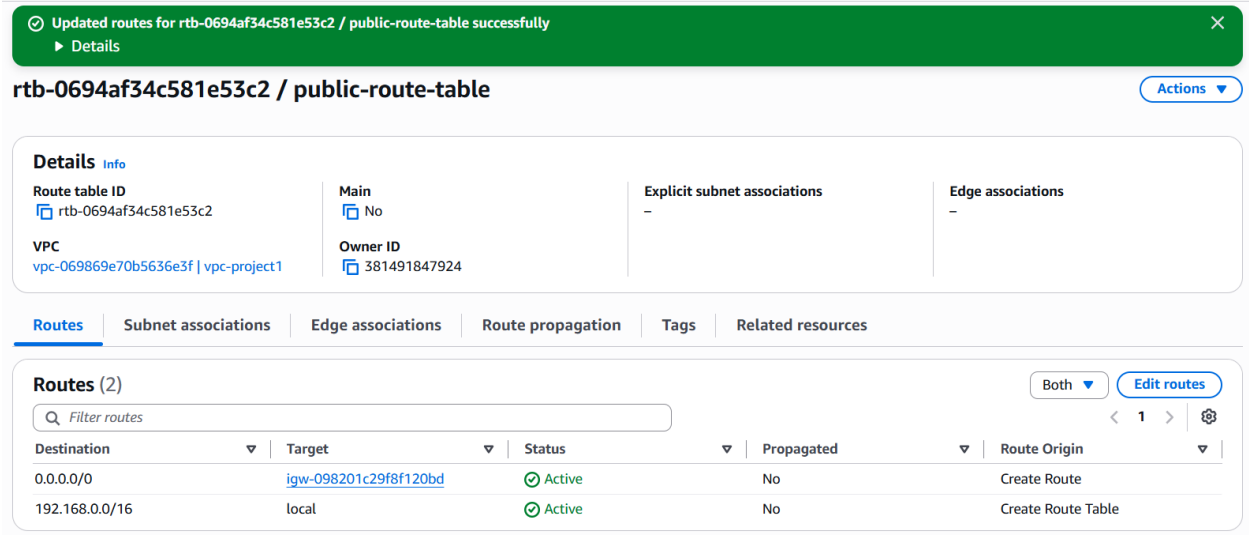
Create internet gateway and attach to VPC



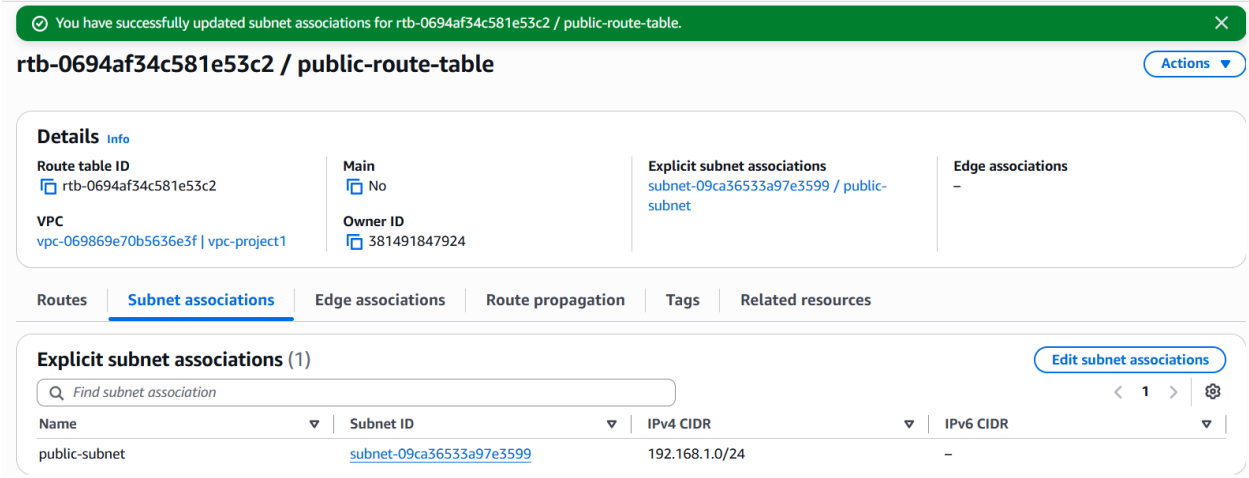
Create public route table



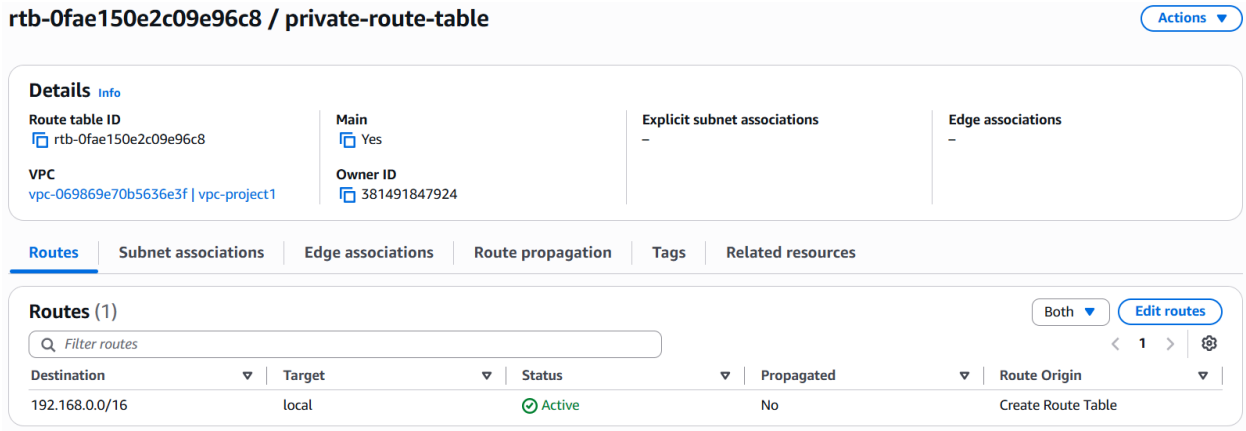
Add internet gateway to public route table



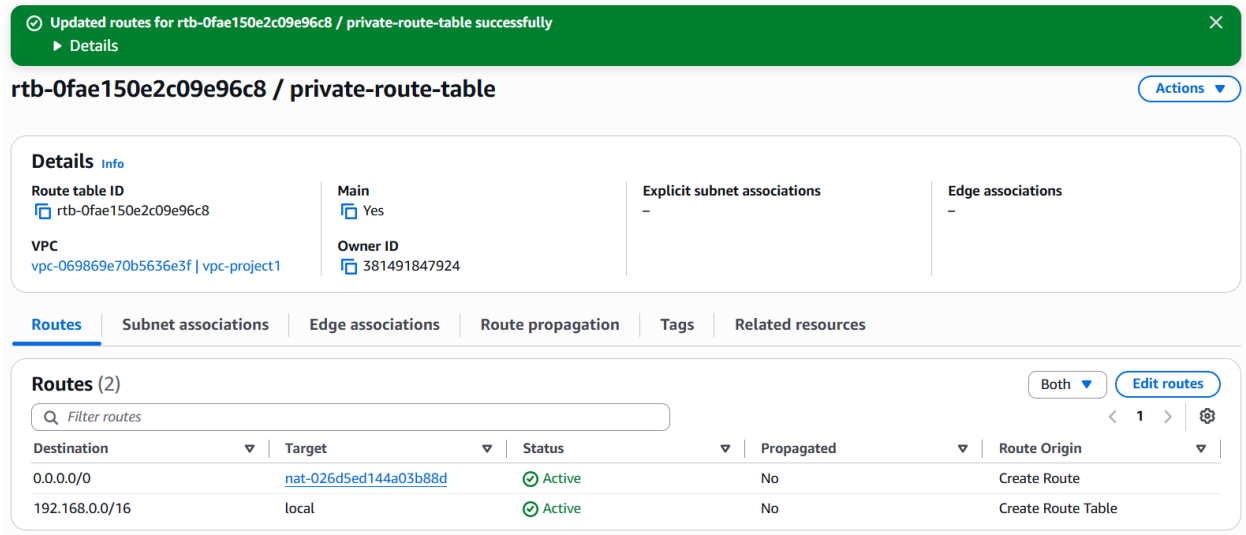
Associate public subnet to public route table



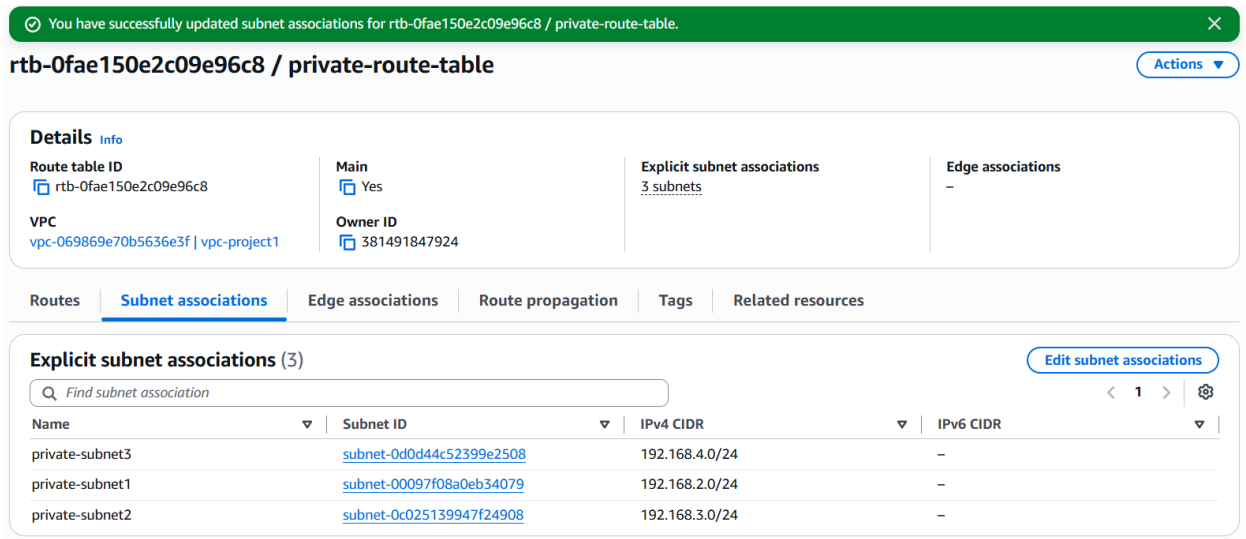
Create private route table



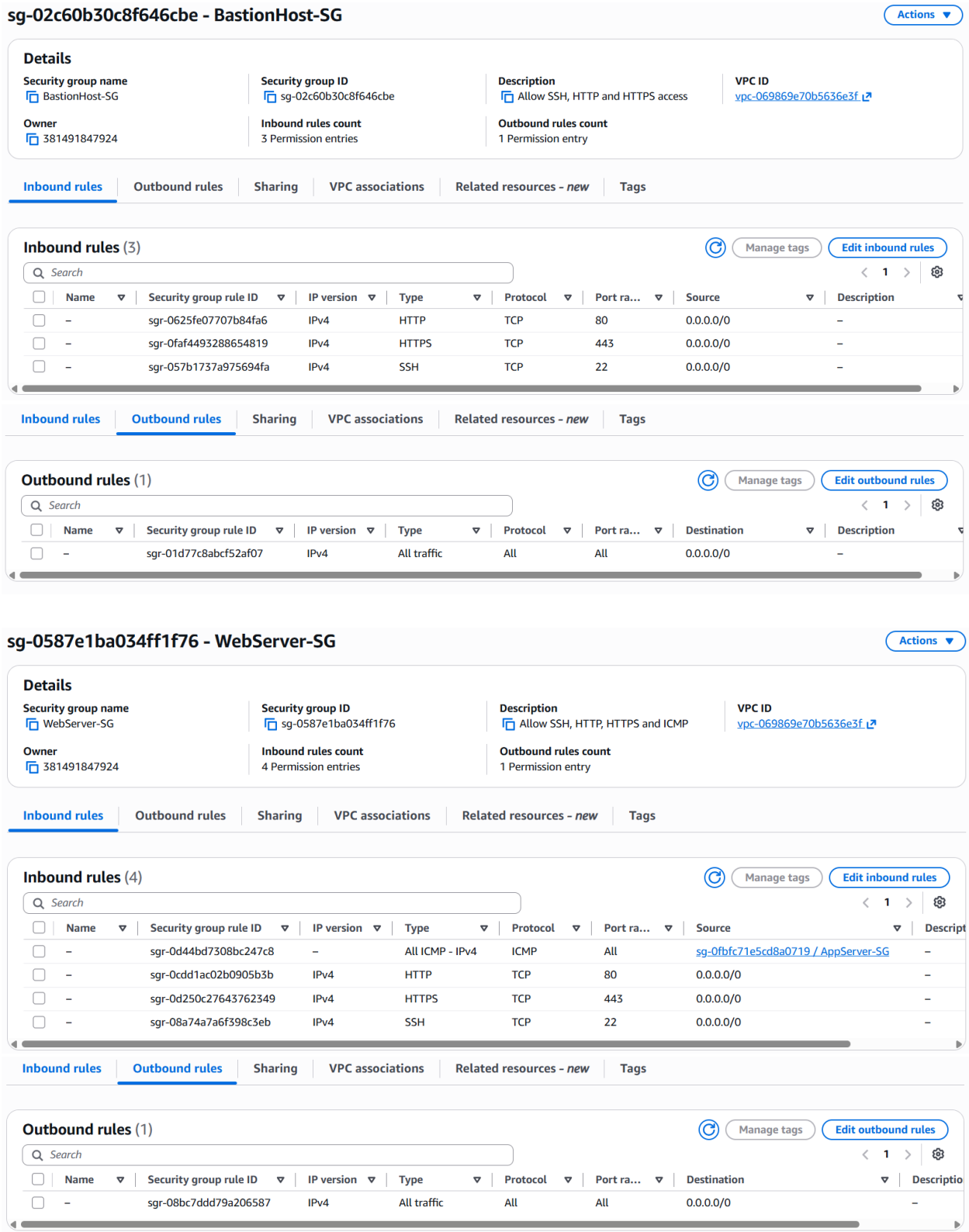
Add NAT gateway to private route table



Associate private subnets to private route table



Create security groups for Bastion Host, Web Server, App Server and Database Server



sg-02c60b30c8f646cbe - BastionHost-SG

Actions

Details

Security group name	Security group ID	Description	VPC ID
BastionHost-SG	sg-02c60b30c8f646cbe	Allow SSH, HTTP and HTTPS access	vpc-069869e70b5636e3f

Owner	Inbound rules count	Outbound rules count
381491847924	3 Permission entries	1 Permission entry

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Inbound rules (3)

Manage tags | Edit inbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port ra...	Source	Description
–	sgr-0625fe07707b84fa6	IPv4	HTTP	TCP	80	0.0.0.0/0	–
–	sgr-0faf4493288654819	IPv4	HTTPS	TCP	443	0.0.0.0/0	–
–	sgr-057b1737a975694fa	IPv4	SSH	TCP	22	0.0.0.0/0	–

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Outbound rules (1)

Manage tags | Edit outbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port ra...	Destination	Description
–	sgr-01d77c8abcf52af07	IPv4	All traffic	All	All	0.0.0.0/0	–

sg-0587e1ba034ff1f76 - WebServer-SG

Actions

Details

Security group name	Security group ID	Description	VPC ID
WebServer-SG	sg-0587e1ba034ff1f76	Allow SSH, HTTP, HTTPS and ICMP	vpc-069869e70b5636e3f

Owner	Inbound rules count	Outbound rules count
381491847924	4 Permission entries	1 Permission entry

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Inbound rules (4)

Manage tags | Edit inbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port ra...	Source	Descript
–	sgr-0d44bd7308bc247c8	–	All ICMP - IPv4	ICMP	All	sg-0fbfc71e5cd8a0719 / AppServer-SG	–
–	sgr-0cdd1ac02b0905b3b	IPv4	HTTP	TCP	80	0.0.0.0/0	–
–	sgr-0d250c27643762349	IPv4	HTTPS	TCP	443	0.0.0.0/0	–
–	sgr-08a74a7a6f398c3eb	IPv4	SSH	TCP	22	0.0.0.0/0	–

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Outbound rules (1)

Manage tags | Edit outbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port ra...	Destination	Descriptio
–	sgr-08bc7ddd79a206587	IPv4	All traffic	All	All	0.0.0.0/0	–

sg-0fbfc71e5cd8a0719 - AppServer-SG

Actions

Details

Security group name	Security group ID	Description	VPC ID
AppServer-SG	sg-0fbfc71e5cd8a0719	Allow SSH, HTTP and HTTPS	vpc-069869e70b5636e3f

Owner	Inbound rules count	Outbound rules count
381491847924	3 Permission entries	1 Permission entry

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Inbound rules (3)

Manage tags | Edit inbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port ra...	Source	Descriptio
–	sgr-0bcb9e833ff272d20	–	HTTPS	TCP	443	sg-0587e1ba034ff1f76 / WebServer-SG	–
–	sgr-06120ab48098bd9de	–	SSH	TCP	22	sg-02c60b30c8f646cbe / BastionHost-SG	–
–	sgr-08b0d447b656f566d	–	HTTP	TCP	80	sg-0587e1ba034ff1f76 / WebServer-SG	–

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Outbound rules (1)

Manage tags | Edit outbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port ra...	Destination	Descriptio
–	sgr-0fe0b27c8e3e3978f	IPv4	All traffic	All	All	0.0.0.0/0	–

sg-04a630bef0fca5ab0 - DatabaseServer-SG

Actions

Details

Security group name	Security group ID	Description	VPC ID
DatabaseServer-SG	sg-04a630bef0fca5ab0	Allow MySQL access	vpc-069869e70b5636e3f

Owner	Inbound rules count	Outbound rules count
381491847924	2 Permission entries	1 Permission entry

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Inbound rules (2)

Manage tags | Edit inbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port ra...	Source	Descriptio
–	sgr-0a496cea8d2b132df	–	MYSQL/Aurora	TCP	3306	sg-02c60b30c8f646cbe / BastionHost-SG	–
–	sgr-02b7129d56db8b22f	–	MYSQL/Aurora	TCP	3306	sg-0fbfc71e5cd8a0719 / AppServer-SG	–

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

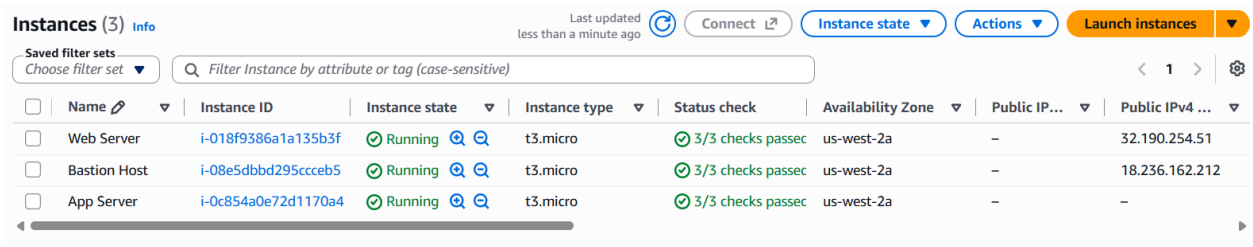
Outbound rules (1)

Manage tags | Edit outbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port ra...	Destination	Descriptio
–	sgr-0fe37bf68299af5ba	IPv4	All traffic	All	All	0.0.0.0/0	–

Create Bastion Host, App Server and Web Server instances

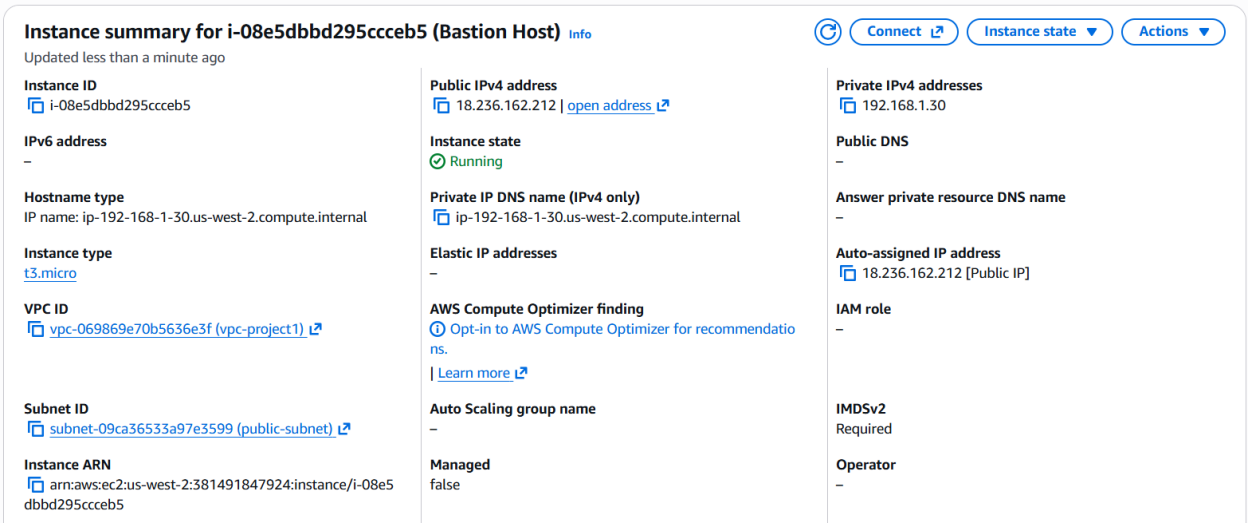


Instances (3) Info

Last updated less than a minute ago | Connect | Instance state | Actions | Launch instances

Saved filter sets: Choose filter set | Filter Instance by attribute or tag (case-sensitive) | 1

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IP...	Public IPv4 ...
Web Server	i-018f9386a1a135b3f	Running	t3.micro	3/3 checks passec	us-west-2a	–	32.190.254.51
Bastion Host	i-08e5dbbd295ccceb5	Running	t3.micro	3/3 checks passec	us-west-2a	–	18.236.162.212
App Server	i-0c854a0e72d1170a4	Running	t3.micro	3/3 checks passec	us-west-2a	–	–



Instance summary for i-08e5dbbd295ccceb5 (Bastion Host) Info

Connect | Instance state | Actions

Updated less than a minute ago

Instance ID
i-08e5dbbd295ccceb5

IPv6 address
–

Hostname type
IP name: ip-192-168-1-30.us-west-2.compute.internal

Instance type
t3.micro

VPC ID
vpc-069869e70b5636e3f (vpc-project1)

Subnet ID
subnet-09ca36533a97e3599 (public-subnet)

Instance ARN
arn:aws:ec2:us-west-2:381491847924:instance/i-08e5dbbd295ccceb5

Public IPv4 address
18.236.162.212 | open address

Instance state
Running

Private IP DNS name (IPv4 only)
ip-192-168-1-30.us-west-2.compute.internal

Elastic IP addresses
–

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling group name
–

Managed
false

Private IPv4 addresses
192.168.1.30

Public DNS
–

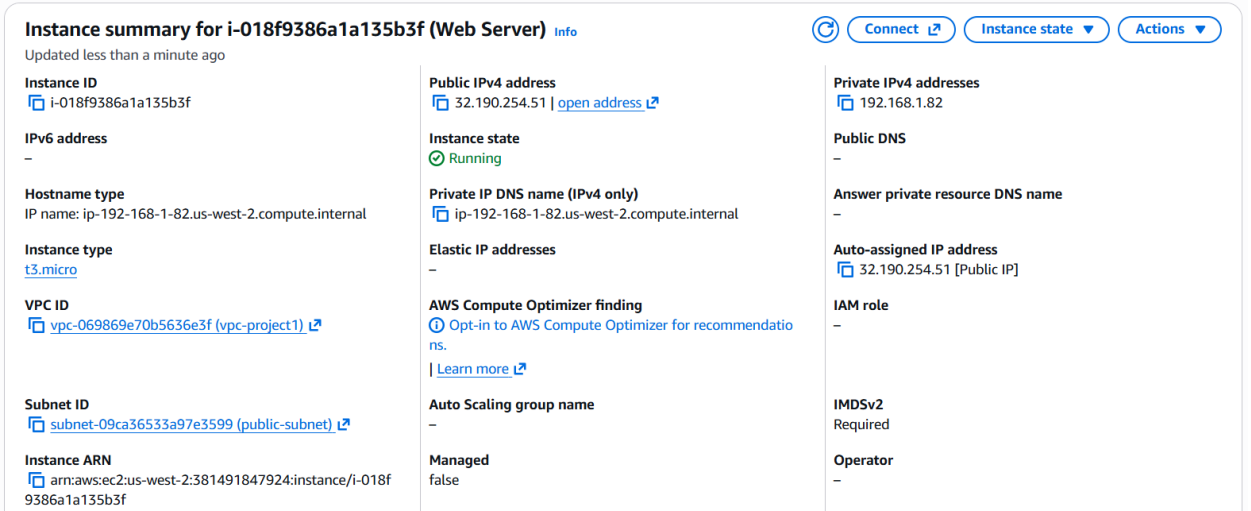
Answer private resource DNS name
–

Auto-assigned IP address
18.236.162.212 [Public IP]

IAM role
–

IMDSv2
Required

Operator
–



Instance summary for i-018f9386a1a135b3f (Web Server) Info

Connect | Instance state | Actions

Updated less than a minute ago

Instance ID
i-018f9386a1a135b3f

IPv6 address
–

Hostname type
IP name: ip-192-168-1-82.us-west-2.compute.internal

Instance type
t3.micro

VPC ID
vpc-069869e70b5636e3f (vpc-project1)

Subnet ID
subnet-09ca36533a97e3599 (public-subnet)

Instance ARN
arn:aws:ec2:us-west-2:381491847924:instance/i-018f9386a1a135b3f

Public IPv4 address
32.190.254.51 | open address

Instance state
Running

Private IP DNS name (IPv4 only)
ip-192-168-1-82.us-west-2.compute.internal

Elastic IP addresses
–

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling group name
–

Managed
false

Private IPv4 addresses
192.168.1.82

Public DNS
–

Answer private resource DNS name
–

Auto-assigned IP address
32.190.254.51 [Public IP]

IAM role
–

IMDSv2
Required

Operator
–

Instance summary for i-0c854a0e72d1170a4 (App Server) Info

Connect | Instance state | Actions

Updated less than a minute ago

Instance ID i-0c854a0e72d1170a4	**Public IPv4 address** –	**Private IPv4 addresses** 192.168.2.116
IPv6 address –	**Instance state** Running	**Public DNS** –
Hostname type IP name: ip-192-168-2-116.us-west-2.compute.internal	**Private IP DNS name (IPv4 only)** ip-192-168-2-116.us-west-2.compute.internal	**Answer private resource DNS name** –
Instance type t3.micro	**Elastic IP addresses** –	**Auto-assigned IP address** –
VPC ID vpc-069869e70b5636e3f (vpc-project1)	**AWS Compute Optimizer finding** Opt-in to AWS Compute Optimizer for recommendations. | Learn more	**IAM role** –
Subnet ID subnet-00097f08a0eb34079 (private-subnet1)	**Auto Scaling group name** –	**IMDSv2** Required
Instance ARN arn:aws:ec2:us-west-2:381491847924:instance/i-0c854a0e72d1170a4	**Managed** false	**Operator** –

Create Database Subnet Group

Successfully created db-subnet-group. View subnet group

Subnet groups (1)

Edit | Delete | Create DB subnet group

Filter by subnet group

1

Name	Description	Status	VPC
db-subnet-group	DB Subnet Group	Complete	vpc-069869e70b5636e3f

db-subnet-group

Subnet group details

VPC ID
vpc-069869e70b5636e3f

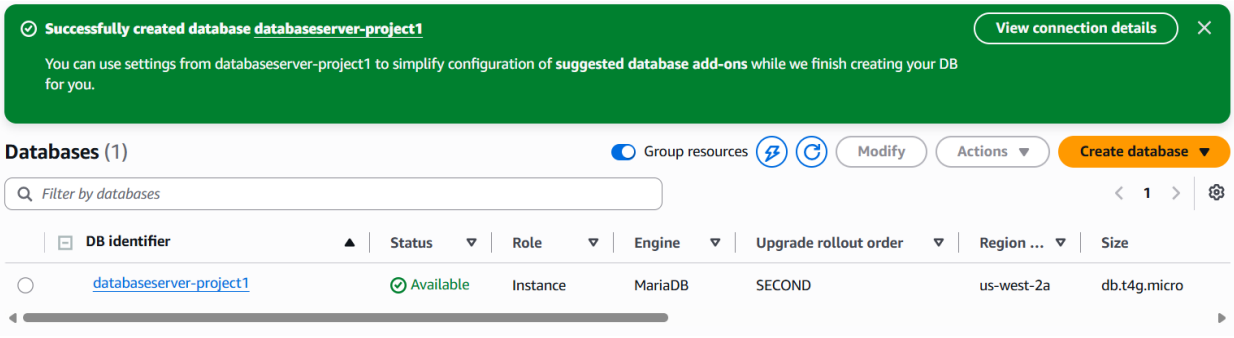
ARN
arn:aws:rds:us-west-2:381491847924:subgrp:db-subnet-group

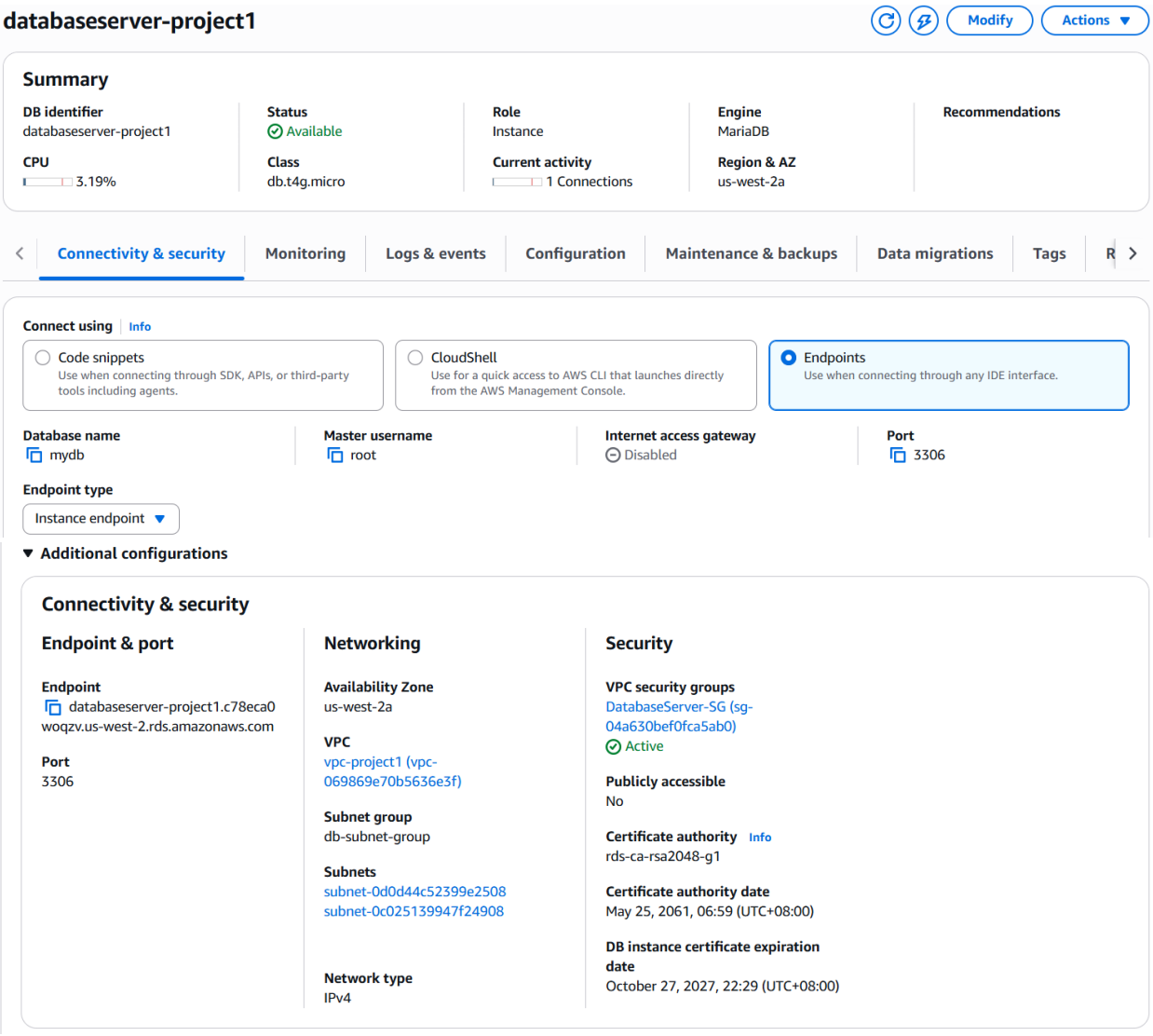
Supported network types
IPv4

Description
DB Subnet Group

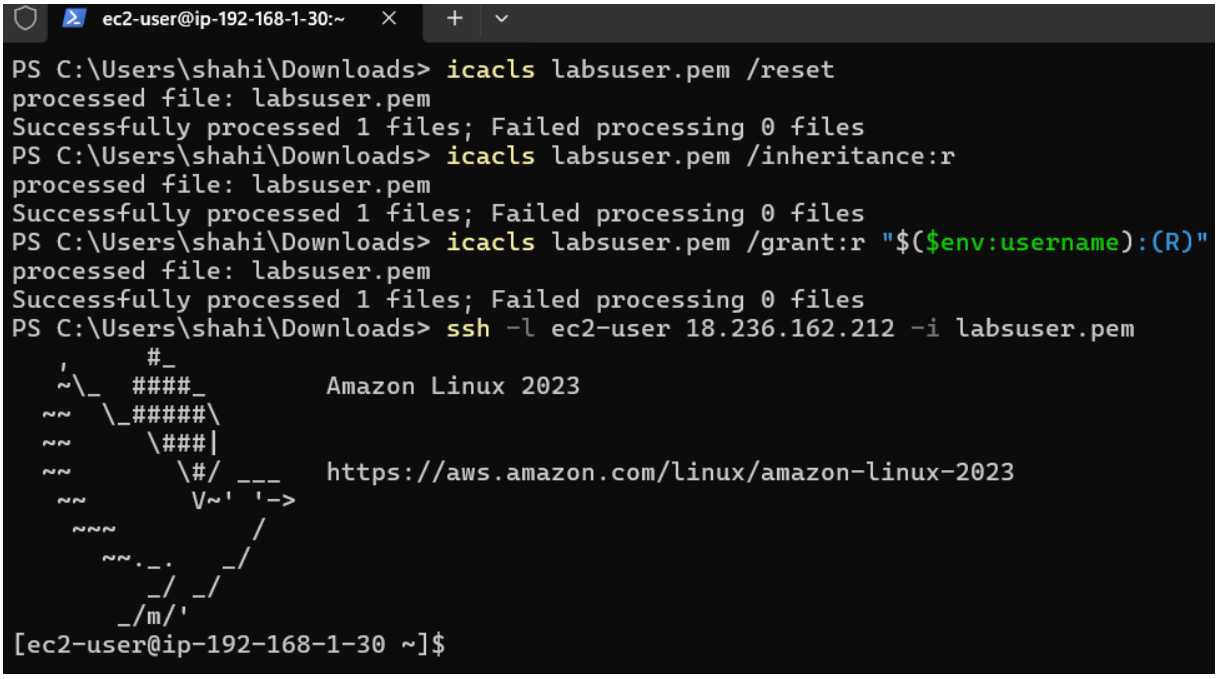
Subnets (2)

Availability zone	Subnet name	Subnet ID	CIDR block
us-west-2b	private-subnet3	subnet-0d0d44c52399e2508	192.168.4.0/24
us-west-2a	private-subnet2	subnet-0c025139947f24908	192.168.3.0/24

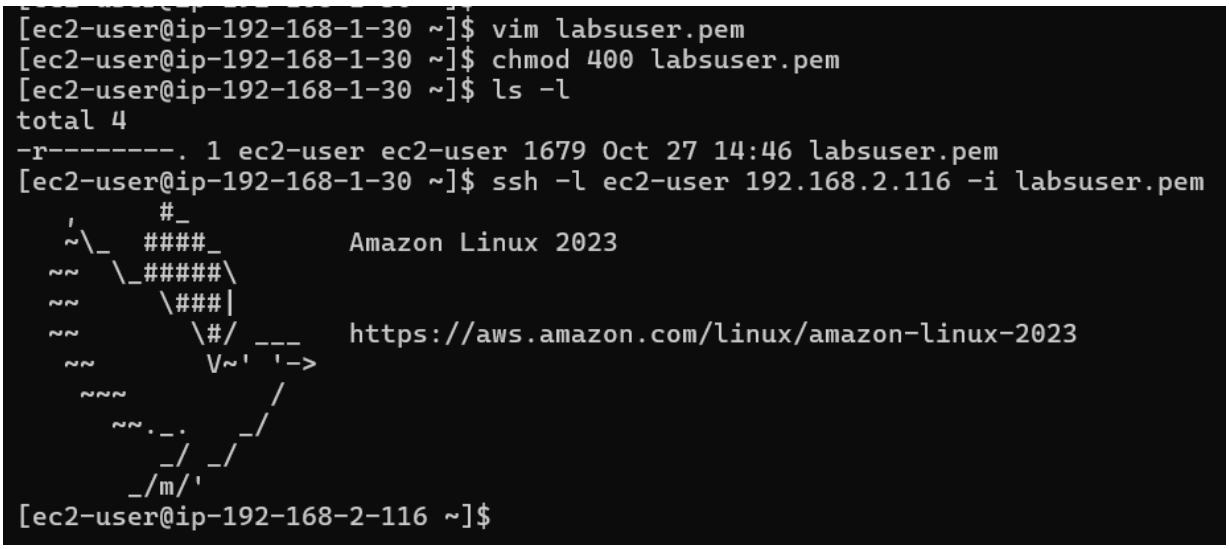

Successfully created database databaseserver-project1
View connection details
You can use settings from databaseserver-project1 to simplify configuration of suggested database add-ons while we finish creating your DB for you.
Databases (1)
Group resources
Modify
Actions
Create database
Filter by databases
1
DB identifier
Status
Role
Engine
Upgrade rollout order
Region ...
Size
databaseserver-project1
Available
Instance
MariaDB
SECOND
us-west-2a
db.t4g.micro


databaseserver-project1
Modify
Actions
Summary
DB identifier
databaseserver-project1
Status
Available
Role
Instance
Engine
MariaDB
Recommendations
CPU
3.19%
Class
db.t4g.micro
Current activity
1 Connections
Region & AZ
us-west-2a
Connectivity & security
Monitoring
Logs & events
Configuration
Maintenance & backups
Data migrations
Tags
Connect using
Info
Code snippets
Use when connecting through SDK, APIs, or third-party tools including agents.
CloudShell
Use for a quick access to AWS CLI that launches directly from the AWS Management Console.
Endpoints
Use when connecting through any IDE interface.
Database name
mydb
Master username
root
Internet access gateway
Disabled
Port
3306
Endpoint type
Instance endpoint
Additional configurations
Connectivity & security
Endpoint & port
Endpoint
databaseserver-project1.c78eca0woqzv.us-west-2.rds.amazonaws.com
Port
3306
Networking
Availability Zone
us-west-2a
VPC
vpc-project1 (vpc-069869e70b5636e3f)
Subnet group
db-subnet-group
Subnets
subnet-0d0d44c52399e2508
subnet-0c025139947f24908
Network type
IPv4
Security
VPC security groups
DatabaseServer-SG (sg-04a630bef0fca5ab0)
Active
Publicly accessible
No
Certificate authority
Info
rds-ca-rsa2048-g1
Certificate authority date
May 25, 2061, 06:59 (UTC+08:00)
DB instance certificate expiration date
October 27, 2027, 22:29 (UTC+08:00)

Connect to Bastion Host



ssh into app server thru bastion host



Ping to web server

```
[ec2-user@ip-192-168-2-116 ~]$ ping 192.168.1.82
PING 192.168.1.82 (192.168.1.82) 56(84) bytes of data.
64 bytes from 192.168.1.82: icmp_seq=1 ttl=127 time=0.614 ms
64 bytes from 192.168.1.82: icmp_seq=2 ttl=127 time=0.217 ms
64 bytes from 192.168.1.82: icmp_seq=3 ttl=127 time=0.238 ms
64 bytes from 192.168.1.82: icmp_seq=4 ttl=127 time=0.229 ms
64 bytes from 192.168.1.82: icmp_seq=5 ttl=127 time=0.243 ms
64 bytes from 192.168.1.82: icmp_seq=6 ttl=127 time=0.231 ms
^C
--- 192.168.1.82 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5172ms
rtt min/avg/max/mdev = 0.217/0.295/0.614/0.142 ms
[ec2-user@ip-192-168-2-116 ~]$
```

Connecting to database

```
[ec2-user@ip-192-168-2-116 ~]$ mysql -h databaseserver-project1.c78eca0woqzv.us-west-2.rds.amazonaws.com -P 3306 -u root
ERROR 3159 (08004): Connections using insecure transport are prohibited while --require_secure_transport=ON.
[ec2-user@ip-192-168-2-116 ~]$ mysql -h databaseserver-project1.c78eca0woqzv.us-west-2.rds.amazonaws.com -P 3306 -u root -p --ssl
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 79
Server version: 11.8.9-MariaDB managed by https://aws.amazon.com/rds/

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> SHOW Databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| innodb             |
| mydb               |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
6 rows in set (0.003 sec)

MariaDB [(none)]>
```